Homework 6

Due October 9, 2025.

Problem 1.

Because it is often needed in string theory, supersymmetry, and supergravity, and also because this is an excellent exercise to prepare for the midterm exam, we study the Dirac matrices and spinors in $d = 3 + 1$, $d = 4 + 0$, $d = 9 + 1$, $d = 10 + 0$, $d = 10 + 1$, $d = 11 + 0$.

a) Begin with $d = 3 + 1$. What are the reality properties of γ^μ? Do Majorana or symplectic Majorana spinors exist? Are the charge conjugation matrices symmetric or antisymmetric? Are they block-diagonal or block off-diagonal? What are their spinor indices?

b) Repeat for $d = 4 + 0$.

c) Repeat for $d = 9 + 1$.

d) Repeat for $d = 10 + 0$.

e) Repeat for $d = 10 + 1$.

f) Repeat for $d = 11 + 0$.

g) In which cases do Majorana-Weyl spinors exist, and in which cases do symplectic Majorana-Weyl spinors exist?